

MANOJ G

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Education

PSG Institute of Technology and Applied Research, Coimbatore

2023 – 2027

– B.Tech in Computer Science & Business Systems

CGPA: 8.31 / 10

Experience

Froze Communication

Feb 2026 – Present

Software Engineer Intern

Coimbatore

- Architected and deployed a **multi-tenant security monitoring platform** with strict tenant isolation and RBAC, enabling scalable multi-organization monitoring from a single deployment.
- Engineered a **Windows endpoint agent** with concurrent workers (event collection, session tracking, firewall sync, auto-update), supporting resilient batched ingestion with deduplication to eliminate duplicate events across retries and restarts.
- Designed a near real-time **EDR-style threat detection and mitigation pipeline** with auto-blocking, ~30s firewall sync, and SSE-based live alerts, reducing manual SOC effort and response latency.

Projects

0xMemory

Python, FastAPI, ChromaDB, MCP | 🐙

- Built a **local-first memory layer for AI agents** using the Model Context Protocol (MCP), enabling persistent, project-scoped context across sessions without cloud dependency.
- Designed a **dual-storage architecture** syncing Markdown files with a ChromaDB vector index, enabling hybrid keyword + semantic search with 100% local data privacy.
- Implemented an **LLM-powered knowledge extraction pipeline** to convert raw chat logs into structured atomic facts, with on-demand recall to optimize token usage and scale memory efficiently.

0xLynk

Node.js, WebRTC, WebSocket, Docker, Playwright | 🐙

- Built a **browser-native P2P file transfer system** using WebRTC DataChannels, ensuring file data never passes through the server and enabling privacy-preserving transfers.
- Engineered **reliable and secure data transfer** with SHA-256 integrity checks, chunk-based resend recovery, per-IP rate limiting, session TTL, and optional passphrase protection.
- Containerized the system using **Docker** with a one-click deployment setup and implemented Playwright-based end-to-end tests to validate cross-browser file transfer reliability.

0xShelf

React, Node.js, MongoDB, Firebase, Redux | 🐙

- Developed a **full-stack MERN application** with modular frontend/backend architecture, featuring a responsive React UI and RESTful API with strict schema validation.
- Implemented **secure authentication and authorization** using Firebase Auth, JWT tokens, and bcrypt hashing, supporting role-based access for Admin and User workflows.
- Built a **real-time inventory and order management system** with analytics dashboard, paginated APIs, and consistent error handling for seamless frontend integration.

Technical Skills

Languages: Python, JavaScript, Java, SQL, Bash

Frameworks & Libraries: FastAPI, Node.js, Express.js, React, Tailwind CSS

Databases: MongoDB, PostgreSQL, ChromaDB

Developer Tools: Git, Docker, Linux, Postman, CI/CD, GitHub Actions

AI/LLM: MCP (Model Context Protocol), AI Agents, LLMs, Vector Embeddings, Prompt/Context Engineering

Areas of Interest

Operating Systems | DBMS | Computer Networks | Object Oriented Programming